

2.1

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$ $\sum y_i = 10$ $\sum (x_i - \bar{x}) = 0$ $\sum (x_i - \bar{x})^2 = 10$ $\sum (y_i - \bar{y}) = 0$					$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

$$a. \bar{x} = \frac{\sum x_i}{N} = 1, \bar{y} = \frac{\sum y_i}{N} = 2$$

$$b. b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8, b_1 = \bar{y} - b_2 \bar{x} = 1.2 \quad \text{where } \bar{x} = \frac{\sum x_i}{N}, \bar{y} = \frac{\sum y_i}{N} \text{ are the sample means of}$$

observations on y and x

$$c. \sum x_i^2 = 15, \sum x_i y_i = 18 \quad \sum (x_i - \bar{x})^2 = \sum x_i^2 - N \bar{x}^2 = 15 - 5 \cdot 1^2 = 10$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \cdot 1 \cdot 2 = 8$$

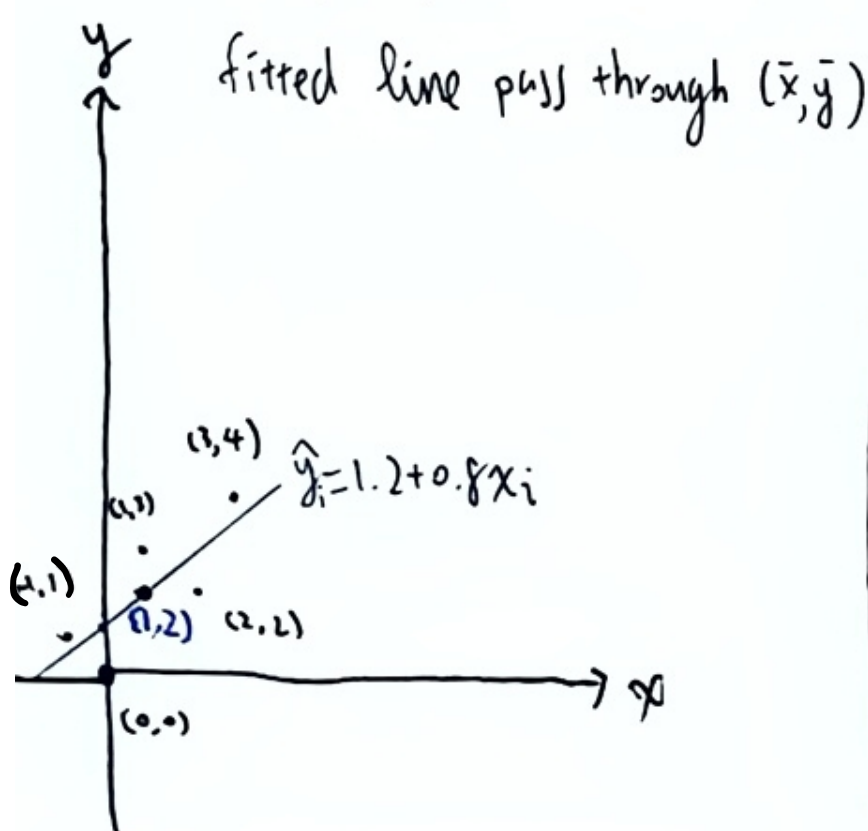
x_i	y_i	$\hat{y}_i$	e_i	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$ $\sum y_i = 10$ $\sum \hat{y}_i = 10$ $\sum e_i = 0$ $\sum \hat{e}_i^2 = 3.6$ $\sum x_i \hat{e}_i = 0$					$\sum (y_i - \bar{y})^2 = 2.5$ $\sum (x_i - \bar{x})^2 = 10$
					$\sum x_i y_i = 18$ $\sum x_i \hat{y}_i = 10$

$$s_y^2 = \frac{\sum (y_i - \bar{y})^2}{N-1} = 2.5 \quad s_x^2 = \frac{\sum (x_i - \bar{x})^2}{N-1} = 2.5$$

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N-1} = 2 \quad r_{xy} = \frac{s_{xy}}{s_x s_y} = 0.8$$

$$(V_x = 100 \frac{s_x}{\bar{x}} = 158.11\% \text{ Medium of } x = 1$$

$$e.f. (\bar{x}, \bar{y}) = (1, 2)$$



$$g. \bar{y} = b_1 + b_2 \bar{x}$$

$$2 = 1.2 + 0.8 \cdot 1$$

$$i. \hat{\sigma}^2 = \frac{\sum (y_i - b_1 - b_2 x_i)^2}{N-2} = \frac{3.6}{3} = 1.2$$

$$h. \hat{y} = \bar{y} \Rightarrow \frac{\sum \hat{y}_i}{N} = \bar{y}$$

$$\frac{10}{5} = 2$$

$$j. \hat{\text{var}}(b_2 | x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = 0.12$$

$$se(b_2) = \sqrt{0.12} = 0.3464$$